Marmara University Electrical-Electronics Eng. Dept.

CSE2003 Digital Design Laboratory Assignment #4

The studies will be uploaded to the system before due date (next week). **Each group must present their own study. The group members have to work together and each member must know each step of the study.**

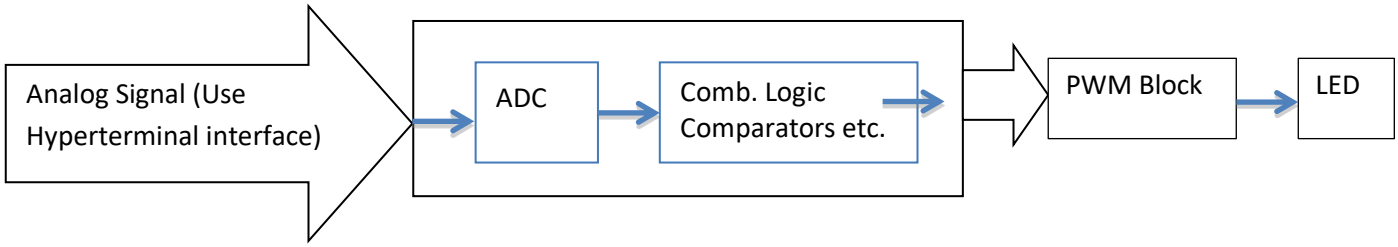
For this laboratory assignment, you will use Altera Quartus II software and DE-115 FPGA kit and design a hardware floating point multiplier that has the Input/output defiitions below.

The steps you need to follow: 1. You should design the circuit, **2.**write the verilog HDL code fort he designed circuit, **3.** Check if the designed hardware is correctly coded by using RTL viewer. Don't forget: your circuit may look different then yours since the compiler can make some optimizations. Please take this information into account when you compare the RTL viewer circuit and yours. **4.** Verify your design by using simulation tool (you need to check the inputs and outputs). **5.** implement and RUN your code using Altera DE-115 FPGA kit. Use the remote desktop connection (**Remote desktop connection ID**:256 521 711 **password**: dijitallab2021).

You need to upload the following files before due date:

1. A short video (<60 secs.) that shows all steps of the study. Especially, the implementation part must be included.
2. The report (in PDF format): 1. What is the purpose of the study? 2. What are the design details with necessary explenations. 3. The design and implementation steps (all steps with screenshots etc.). 4. The result and coclusion: did you reach the purposed result? If the answer is no, why? if it is yes, how?

For this laboratory assignment, you will use Altera Quartus II software and DE-115 FPGA kit and design an analog signal comparator that has the Input/output definitions below.



- The analog signal, which is applied to ADC input is changing from 0 to 3V. The ADC is an 8-bit device, which has the output pins connected to FPGA.
- Combinational logic will be designed by using three 8-bit comparators that include 4x1 multiplexers.
- The circuit should measure the analog input and the duty cycle of the PWM generated by FPGA device is changing according to analog input value. You can see the duty cycle values (for the input voltages) in the table below:

$0 \leq Vin < 0.5 \rightarrow$ duty cycle=0%

$0.5 \leq Vin < 1.5 \rightarrow$ cycle=30%

$1.5 \leq Vin < 2.5 \rightarrow$ cycle=60%

$2.5 \leq Vin < 3.0 \rightarrow$ cycle=100%

You can choose the PWM period value. Please follow the steps given below:

1. Design a 4x1 multiplexer module,

2. Design the 8-bit comparator module (Use the multiplexer modules),

3. Use the comparator modules and generate the proper PWM signals and derive the FPGA code.

4. Synthesize the circuit, program the FPGA, and verify its operation.

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